

Exploring DPO's Implicit Reward

Preliminary Results & Challenges — Project Update
Graduate Reinforcement Learning | 14 August 2026

Group Members

#	Name	Student ID	Email
1	«FILL IN»	«FILL IN»	shamima2hossain@gmail.com
2	«FILL IN»	«FILL IN»	«FILL IN»
3	«FILL IN»	«FILL IN»	«FILL IN»

Replace «FILL IN» before submitting.

1 Problem Definition and MDP Specification

Direct Preference Optimization (DPO) reparameterizes the RLHF reward in terms of the policy. Its central claim is that a DPO-trained language model *is* a reward model via the **implicit reward** $\hat{r}(x, y) = \beta \cdot \log[\pi_\theta(y|x) / \pi_{\text{ref}}(y|x)]$. This equivalence is exact only at the optimum and on the training distribution. We ask whether the implicit reward **generalizes as a reward model** as well as an explicitly-trained Bradley–Terry (BT) reward model, under matched data and compute.

1.1 Token-level MDP

Autoregressive generation is a finite-horizon, deterministic MDP $M = (S, A, P, R, \rho_0, \gamma)$:

- **State** $s_t = (x, y_{\leq t})$: the prompt x together with the response tokens generated so far.
- **Action** $a_t = y_t \in V$: emit the next token from the vocabulary V .
- **Transition** $P(s_{t+1}|s_t, a_t)$: deterministic append, $s_{t+1} = (x, y_{\leq t+1})$.
- **Reward** R : sparse and terminal — a scalar $r(x, y)$ delivered at the end-of-sequence token; the KL-regularized RLHF objective adds a per-token shaping term $-\beta \cdot \log[\pi_\theta / \pi_{\text{ref}}]$.
- **Initial state** ρ_0 : a prompt $x \sim D$ with empty response.
- **Horizon / discount**: episodic with $H = \max$ new tokens, $\gamma = 1$. The policy is the LM $\pi_\theta(a_t|s_t)$.

1.2 Contextual-bandit abstraction (DPO's view)

DPO collapses the horizon: the context is the prompt $x \sim D$, the action is the *entire* response $y \sim \pi(\cdot|x)$, and the reward is the terminal scalar $r(x, y)$. The KL-regularized objective is

$$\max_{\pi} E_{x \sim D, y \sim \pi(\cdot|x)} [r(x, y)] - \beta \cdot D_{KL}(\pi(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)).$$

This KL-constrained problem has the closed-form optimum

$$\pi^*(y|x) = (1/Z(x)) \cdot \pi_{\text{ref}}(y|x) \cdot \exp(r(x, y)/\beta),$$

which, solved for the reward, gives the reparameterization $r(x, y) = \beta \cdot \log[\pi^*(y|x)/\pi_{\text{ref}}(y|x)] + \beta \cdot \log Z(x)$. The partition term $\beta \cdot \log Z(x)$ depends on x only, so it cancels in any same-prompt pairwise comparison.

1.3 Preference model, DPO loss, and the two scorers

Preferences follow the Bradley–Terry model $P(y_w > y_l | x) = \sigma(r(x, y_w) - r(x, y_l))$, where $y_w > y_l$ denotes that y_w is preferred to y_l . Substituting the reparameterized reward ($Z(x)$ cancels) yields the **DPO loss**

$$L_{DPO} = -E_{(x, y_w, y_l)} \log \sigma(\beta \log[\pi_\theta(y_w|x)/\pi_{ref}(y_w|x)] - \beta \log[\pi_\theta(y_l|x)/\pi_{ref}(y_l|x)]).$$

Implicit reward (object of study). $\hat{r}_\theta(x, y) = \beta \log[\pi_\theta(y|x)/\pi_{ref}(y|x)] = \beta \sum_t (\log \pi_\theta(y_t|x, y_{<t}) - \log \pi_{ref}(y_t|x, y_{<t}))$.

Explicit reward (baseline). A scalar head $r_\phi(x, y)$ trained with the BT loss $L_{RM} = -E \log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$ on the identical pairs, from the identical SFT initialization.

Evaluation as a preference classifier. For a held-out pair, predict $y_w > y_l$ iff $r(x, y_w) > r(x, y_l)$; the primary metric is **pairwise accuracy**. Note ranking is invariant to β (a positive scalar), so β affects calibration/ECE, not accuracy.

Research question & hypotheses. Under matched data/compute, how does held-out preference accuracy of $\hat{r}$ compare to r_ϕ , ID and under shift? **H1:** explicit > implicit on held-out ID pairs. **H2:** the gap widens under distribution shift. A null result (implicit $\approx$ explicit) would empirically substantiate DPO's claim at small scale.

2 Benchmark Datasets

Following the instructor's feedback to use benchmark datasets, all data are established public benchmarks:

Dataset	Role in this study
UltraFeedback-binarized (HuggingFaceH4/ultrafeedback_binarized)	Training preference pairs and the in-distribution (ID) held-out test set.
RewardBench (allenai/reward-bench)	The standard reward-model evaluation benchmark; its Chat / Chat-Hard / Safety / Reasoning sub-
Anthropic HH-RLHF (Anthropic/hh-rlhf)	A classic human-preference benchmark used as a cross-dataset OOD set.

3 Initial Baselines: Implementations and Results

3.1 Reference baselines (implemented and run)

These non-neural baselines are the reference floors the neural reward models must beat. They require no GPU and are already evaluated on every benchmark set:

- **Random** — chance (0.500).
- **Pick-longer / Pick-shorter** — predict the longer (resp. shorter) response wins; probes the length confound flagged in the proposal.
- **TF-IDF + Logistic Regression** — a learned lexical scorer: TF-IDF over (prompt, response), trained on UltraFeedback to score a response chosen/rejected, higher-scoring side wins.

Test set	n	Random	Pick-longer	Pick-shorter	TF-IDF+LR
UltraFeedback (ID)	1997	0.500	0.568	0.432	0.618
HH-RLHF harmless (OOD)	2312	0.500	0.445	0.555	0.453
RewardBench: Chat (OOD)	358	0.500	0.804	0.196	0.754
RewardBench: Chat-Hard (OOD)	456	0.500	0.294	0.706	0.382
RewardBench: Safety (OOD)	740	0.500	0.420	0.580	0.450
RewardBench: Reasoning (OOD)	1431	0.500	0.397	0.603	0.282
RewardBench: ALL (OOD)	2985	0.500	0.436	0.564	0.395

Pairwise accuracy. TF-IDF+LR trained on 19,980 UltraFeedback pairs (seed 0). Ties count as 0.5.

3.2 Key preliminary findings

- **Length bias is large and dataset-dependent.** Pick-longer reaches 0.804 on RewardBench-Chat but only 0.294 on the adversarial Chat-Hard (which deliberately inverts length). This validates the proposal's length-controlled metric as essential — a scorer can look strong purely by proxying length.
- **A lexical learner transfers partially.** TF-IDF+LR beats chance in-distribution (0.618) and on easy Chat (0.754), but collapses to near/below chance on Chat-Hard (0.382) and Reasoning (0.282) — quantifying how much of the signal is shallow lexical overlap and setting a floor for the neural models.
- The ID→OOD accuracy drop already visible here previews exactly the H2 comparison we will make between the implicit and explicit rewards.

3.3 Neural conditions (implemented, queued for GPU)

The full training and evaluation pipeline is implemented and syntax-verified; it targets Qwen2.5-0.5B with LoRA via HuggingFace TRL on a single GPU (Colab). Three neural scorers are ready to run:

- **DPO implicit reward** $r^\wedge$ — extracted as the β -scaled response log-ratio (*src/scoring.py*), to be verified against DPOTrainer's own logged rewards.
- **Explicit Bradley–Terry RM** r_ϕ — scalar head, matched to DPO (same SFT init, pairs, epochs, LoRA, batch, LR; see *DESIGN.md*).
- **Base-model log-probability** — a free neural baseline (length-normalized).

Evaluation reports pairwise accuracy (primary), ECE, length-controlled accuracy, Spearman vs gold, and a McNemar paired test between the implicit and explicit scorers (*src/metrics.py*, *src/eval.py*).

4 Challenges

- **Auditable budget-matching.** The comparison is only meaningful if DPO and the RM see identical data/compute. We fix a shared SFT initialization and match epochs, LoRA config, batch, and LR, and log tokens-seen / wall-clock — but defining 'equal compute' across two different losses (DPO's extra reference forward pass) remains a judgment call we must defend.
- **Correct implicit-reward extraction.** The β -scaled, prompt-masked, length-summed log-ratio must exactly match TRL's internal convention; a subtle masking or scaling error would silently bias every downstream number. Mitigation: cross-check against DPOTrainer's logged rewards.
- **Small-model noise.** At 0.5B, reward accuracy can sit close to chance on hard subsets (the TF-IDF collapse on Reasoning previews this), so headline claims need ≥ 3 seeds with confidence intervals; we may escalate to 1.5B if the signal is too noisy.
- **Length as a confound.** Both scorers can proxy length (Section 3.2); isolating true preference signal requires the length-controlled metric and reporting the length-only baseline everywhere.
- **Gold-judge / Spearman set on limited compute.** A larger instruction-tuned judge for scalar gold ratings must be run as a separate quantized/offline pass (or via API) to fit a single consumer GPU alongside training.
- **Toolchain drift.** TRL's DPOTrainer / RewardTrainer APIs shift across releases; versions are pinned in *requirements.txt* and the trainers may need minor version-specific edits on Colab.

5 Next Steps

- Run SFT $\rightarrow$ DPO $\rightarrow$ BT-RM end-to-end at the 8k budget on Colab; produce the first H1/H2 numbers (implicit vs explicit, ID + OOD) with the reference floors above as context.
- Widen to the ablations: $\beta \in \{0.05, 0.1, 0.3, 0.5\}$, budget $\in \{2k, 8k, 32k\}$, and the training-progress checkpoint curve (overoptimization).
- Add the gold-judge OOD set and Spearman correlation; finalize with ≥ 3 seeds and significance tests.

Code, benchmark data pipeline, and reproducible baseline results: github.com/silvererudite/dpo-implicit-reward